Package 'FirnR'

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Title Tools for simulating snow and firn cores including their isotopic composition

Version 0.0.0.9000

Description What the package does (one paragraph).

Depends R (>= 3.3.1)

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Encoding UTF-8

LazyData true

RoxygenNote 5.0.1

R topics documented:

`CIce` *Specific heat capacity of ice*

Description

#Paterson 1994 copied from Goujon, C., J.-M. Barnola, and C. Ritz (2003), Modeling the densification of polar firn including heat diffusion: Application to close-off characteristics and gas isotopic fractionation for Antarctica and Greenland sites, J. Geophys. Res., 108(D24), 4792, doi:10.1029/2002JD003319.

Usage

```
CIce(T)
```

Arguments

T	ice temperature [K]

Value

specific heat capacity of ice J/(kg K)

Author(s)

Thomas Laepple

`Convert2SnowDepth` *Water equivalent depth to snow depth conversion*

Description

Convert data in water equivalent depth to snow depth

Usage

```
Convert2SnowDepth(rho, depth.we, data, dZOut = 0.01)
```

Arguments

rho	vector of density [kg/m^3];
depth.we	vector of depth in m w.e
data	some data vector
dZOut	output resolution (m)

Value

list(depth,data,rho) snow depth and data / rho interpolated to the snow depth

Author(s)

Thomas Laepple

Examples

```
 temp<-DensityHL(rho.surface=340,t.mean=273.15-31.5,bdot=177,depth=0:150)
 depth.we<-Convert2We(temp$rho,temp$depth)
 #Add zero depth
 depth.we<-c(0,depth.we)
 density<-c(temp$rho[1],temp$rho)
 inSnow<-Convert2SnowDepth(rho=density,depth.we=depth.we,data=density)
 plot(temp$depth,temp$rho,type="l",xlab="snow depth",ylab="density")
 lines(inSnow$depth,inSnow$rho,col="red")
legend("topleft",col=c("black","red"),lwd=2,c("original","after forth and back conversion"))
```

Convert2WE *Convert snow depth to water equivalent depth*

Description

Convert snow depth to water equivalent depth by summing up the layers, unfinished

Usage

```
Convert2WE(rho, depth, bCorrectStart = FALSE, accum)
```

Arguments

rho	vector of density in [kg/m^3];
depth	snow depth in m, vector of same length as rho
bCorrectStart	if TRUE, correct for the starting depth using the
accum	annual mean accumulation rate kg/m2/year, only has to be supplied if bCorrectStart=TRUE

Details

often the first density measurement is not directly at the surface. In this case the water equivalent depth is missing the top part, if bCorrectStart is choosen, the mean density of the approx. one year from the exisiting density is estimated and the depth is extrapolated to the surface

Value

vector of depth in m w.e.

Author(s)

Thomas Laepple

Examples

```
temp<-DensityHL(rho.surface=340,t.mean=273.15-31.5,bdot=177,depth=0:150)
Convert2WE(temp$rho,temp$depth)
```

DensityHL *Herron Langway firn density*

Description

Herron Langway firn density model, analytical solution

Usage

```
DensityHL(rho.surface, t.mean, bdot, depth = 0:9000/100)
```

Arguments

`rho.surface`	surface density in kg/m3
`t.mean`	10 firn mean temperature in Kelvin
`bdot`	accumulation rate in kg/m^2/year
`depth`	snow depth in m

Details

based on the Matlab Code from Alan Grinsted #https://www.mathworks.com/matlabcentral/fileexchange/47386-steady-state-snow-and-firn-density-model/content//densitymodel.m

Value

list(depth, rho) with snow depth and density in kg/m^3

Author(s)

Thomas Laepple examples result=DensityHL(rho.surface=340,t.mean=273.15-31.5,bdot=177,depth=0:150) plot(result$rho,result$depth,ylim=c(150,0),xlim=c(200,1000),xlab="firn density",ylab="depth (m)",type="l",lwd=2,mai simulated density")

`DiffusionLength` *water isotope diffusion length in firn*

Description

Diffusion length in firn

Usage

```
DiffusionLength(rho = NULL, z, dz, T, P = 0.65, b = 0.064, dD = FALSE)
```

Arguments

`rho`	firn density *vector* [kg/m^3]; if not provided, a linear density profile is used
`z`	firn depth *vector* [m]; observation points of 'rho', must be of same length as 'rho'
`dz`	depth resolution *vector* [m]; may vary over 'z', must be of same length as 'z'
`T`	temperature [K], either a scalar value or a vector of the same length a rho
`P`	surface pressure [atm]
`b`	annual mean accumulation rate m.w.e per year?
`dD`	*logical*; if true, the diffusion length for d2H is returned, otherwise for d18O

Details

Thomas add details

Value

list(z=z,rho=rho,sigma=sigma), firn depth, density, diffusion length in cm

Author(s)

Thomas Muench modified by Thomas Laepple

Examples

```
depth<-0:150
t.mean<-273.15-30
bdot=200
rho<-DensityHL(rho.surface=340,t.mean=273.15-31.5,bdot=bdot,depth=depth)$rho
result.dO18<-DiffusionLength(rho,depth,dz=1,T=t.mean,b=bdot/1000,dD=FALSE)
result.dD<-DiffusionLength(rho,depth,dz=1,T=t.mean,b=bdot/1000,dD=TRUE)
plot(result.dO18$sigma,result.dO18$z,ylim=c(150,0),xlim=c(0,12),xlab="diffusion length (cm)",ylab="depth (r
lines(result.dD$sigma,result.dD$z,lwd=2,col="red")
legend("topleft",col=c("black","red"),lwd=2,c("d18O","dD"),bty="n")
```

`Diffusivity` *Diffusivity of water isotopes in firn*

Description

Diffusivity of water isotopes in firn

Usage

```
Diffusivity(rho, T, P, dD = FALSE)
```

Arguments

rho	firn density [kg/m^3]
T	annual mean temperature [K]
P	surface pressure [atm]
dD	dD - *logical*; if true, the diffusivity for d2H is returned, otherwise for d18O

Details

Thomas: add details / references

Value

diffusivity in [cm^2/s]

Author(s)

Thomas Muench modified by Thomas Laepple

Examples

```
T<-(-60:0)
plot(T,Diffusivity(rho=340,T=273.15+T,P=0.65,dD=FALSE),type="l",lwd=2,main="temperature dependence of diffu
lines(T,Diffusivity(rho=340,T=273.15+T,P=0.65,dD=TRUE),col="red",lwd=2)
legend("topleft",col=c("black","red"),lwd=2,c("d18O","dD"))
```

`FirnTemperature` *Simulate the Firn temperature*

Description

Simulate the firn gemperature based on the Fourier law of heat diffusion with an input signal given as the superposition of the annual cycle and its second harmonics

Usage

```
FirnTemperature(t, z, core = NULL, A1 = core$A1, A2 = core$A2,
  phi1 = core$phi1, phi2 = core$phi2, T0 = core$T0, kappa)
```

Arguments

`t`	time in years
`z`	snow depth in m, surface = 0, 1 = 1m deep
`core`	list of core parameters or alternativly NULL, in this case A1,A2 ... have to be given explicitly
`A1`	Amplitude of first harmonic (~1/2 of seasonal range)
`A2`	Amplitude of second harmonic
`phi1`	Phase of first harmonic
`phi2`	Phase of second harmonic
`T0`	Mean temperature in Kelvin
`kappa`	Thermal diffusivity of firn m^2/s

Value

vector of temperature at time t at the depth levels z

Author(s)

Thomas Laepple

Examples

```
plot(1:10,type="n",xlim=c(-45,-10),ylim=c(20,0),main="NGRIP")
for (iMonth in 1:12)
lines(FirnTemperature(iMonth/12,(0:1000)/50,A1=16.5,A2=3,phi1=0,phi2=0,T0=-31.5,kappa=KappaFirn(273-31.5,rh
```

KappaFirn *thermal diffusivity of firn*

Usage

```
KappaFirn(T, rho)
```

Arguments

T	firn temperature [K]
rho	firn density [kg/m^3]

Value

thermal diffusivity of firn m^2/s

Author(s)

Thomas Laepple

KappaIce *thermal diffusivity of ice*

Usage

```
KappaIce(T)
```

Arguments

T	ice temperature [K]

Value

thermal diffusivity m^2/s

Author(s)

Thomas Laepple

KFirn *Thermal conductivity of firn*

Description

Thermal conductivity of firn W/(m K) Schwander et al., 1997 copied from Goujon, C., J.-M. Barnola, and C. Ritz (2003), Modeling the densification of polar firn including heat diffusion: Application to close-off charact##' eristics and gas isotopic fractionation for Antarctica and Greenland sites, J. Geophys. Res., 108(D24), 4792, doi:10.1029/2002JD003319.

Usage

```
KFirn(T, rho)
```

Arguments

`T`	firn temperature [K]
`rho`	firn density [kg/m^3]

Value

Thermal conductivity of firn W/(mK)

Author(s)

Thomas Laepple

References

Schwander et al., 1997

KIce *Thermal conductivity of ice W/(mK)*

Description

Thermal conductivity of ice W/(mK) Weller and Schwerdtfeger 1971 copied from Goujon, C., J.-M. Barnola, and C. Ritz (2003), Modeling the densification of polar firn including heat diffusion: Application to close-off characteristics and gas isotopic fractionation for Antarctica and Greenland sites, J. Geophys. Res., 108(D24), 4792, doi:10.1029/2002JD003319.

Usage

```
KIce(T)
```

Arguments

`T`	firn temperature [K]

Value

Thermal conductivity of ice W/(m K)

Author(s)

Thomas Laepple

ParcelTemperature *Temperature of a snow parcel*

Description

Simulate temperature against depth for a snow parcel starting at a particular season based on the Fourier law of heat diffusion. This assumes a constant layer thickness, thus ignoring densification. As the temperature below 5m is close to constant, this approximation should be reasonable.

Usage

```
ParcelTemperature(startTime, depth, core)
```

Arguments

`startTime`	time in the year when the parcel started (years)
`depth`	depth vector (m)
`core`	list containing the core parameters (here A1,A2,phi1,phi2,T0,density and accum) are used

Value

vector of temperatures at the depths given by the depth vector

Author(s)

Thomas Laepple

Examples

```
kohnen<-list(lat=-75.00,lon=0,accum=72,density=345,T0=273.15-44.5,A1=16.7,A2=6.6,phi1=0,phi2=0,P=0.65,name=
depth<-seq(from=0,to=10,by=1/100)
plot(depth,ParcelTemperature(0.5,depth,kohnen),main="Temperature of a parcel at Kohnen",xlab="snow depth",y
```

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